Set-1

MORADABAD INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Sessional Test – 1

Course: B.Tech. **Semester: 3rd**
Session: 2025-2026 **Section: A, B, C, D**
Subject Name: DSTL **Subject Code: BCS303**
Max. Marks: 20 **Time: 60 min**

Q. No.	1	2	3	4	5	6
CO	1	1	2	1	2	3
Bloom's Cognitive Level	K3	K3	K3	K3	K3	K3

Section(A)

Attempt all questions.

Q1. Show that for any two sets A and B: $(A \cup B)' = A' \cap B'$ using the method of mutual subsets. **(2 marks)**

Q2. If R is a relation on the set of integers defined by $R = \{(x, y): (x \cong y) mod\ m\}$, where $\cong$ denotes congruence modulo m, then prove that R is an equivalence relation. **(3 marks)**

Q3. Let A and B be two sets. If $f: A \to B$ is one-to-one and onto, prove that $f^{-1}: B \to A$ is also one-to-one and onto. **(3 marks)**

Section(B)

Attempt all questions. Each question carries 4 marks. **(3*4=12)**

Q4. Show that D_{64}, denoting the set of divisors of 64 ordered by divisibility, forms a lattice $(D_{64}, /)$.

Q5(i). Solve the following Boolean function using a K-map:

$$F(A, B, C, D) = \sum (0,1,2,5,7,8,9,10,13,15).$$

(ii). Also, express the following Boolean expression in conjunctive normal form:

$$(x + y + z)(xy + x' z)'$$

Q6(i). Show that $\left((p \vee q) \wedge (p' \wedge (q' \vee r'))'\right) \vee (p' \wedge q') \vee (p' \vee r)$ is a tautology.

(ii) Show that the statements $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\sim P \wedge \sim Q)$ are equivalent.

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Sessional Test – 2

Course: B.Tech. **Semester: 3rd**

Session: 2025-2026 **Section: A, B, C, D, I, J**

Subject Name: Discrete Structures & Theory of Logic **Subject Code: BCS303**

Max. Marks: 20 **Time: 1 hr**

Q.No.	1	2	3	4	5	6
CO	4	4	5	3	4	5
Bloom's Cognitive level	K3	K4	K5	K3	K3	K3

CO3:	**Employ the rules of propositions and predicate logic to solve the complex and logical problems.**
CO4:	**Explore the concepts of group theory and their applications for solving the advance technological problems.**
CO5:	**Illustrate the principles and concepts of graph theory for solving problems related to computer science.**

Section(A)

Attempt all questions.

Q1. The identity element of any algebraic structure is unique. **(2 marks)**

Q2. Calculate number of generators in a cyclic group of order 8. **(3 marks)**

Q3. State Pigeonhole Principle. If you have 10 black socks and 12 blue socks mixed in a drawer, how many socks must you pick to guarantee a matching pair?. **(3 marks)**

Section(B)

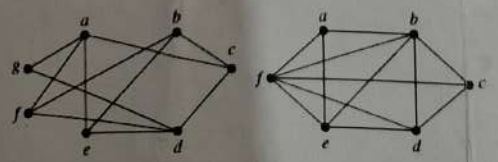
Attempt all questions. Each question carries 4 marks. **(3*4=12)**

Q4. Rewrite the following arguments using quantifiers, variables and predicate symbols

(a) All birds can fly.
(b) Some men are genius.
(c) Some numbers are not rational.
(d) There is a student who likes mathematics but not geography.

Q5. What is Field? Define elementary properties of Field with example.

Q6. Are the graphs G and H bipartite?



G H